

Comprehensive POS System Design Plan

Executive Summary

This document outlines a revised design plan for the Point-of-Sale (POS) system, incorporating real-time order management, multi-channel integration, and robust business logic. It supports dine-in, takeaway, and home delivery operations, with a focus on streamlined workflows and a clear separation of core functionalities. The architecture is refined to align with user-centric pages, treating supporting features like authorization, printing, and tax calculation as integrated functionalities rather than standalone core microservices.

1. System Architecture Overview

The system is designed with a layered architecture, promoting modularity, scalability, and maintainability.

1.1 Core Architecture Components

- **Frontend Layer:** User-facing applications for various touchpoints.
 - **Android POS App:** For order taking, billing, and general POS operations.
 - **Kitchen Display:** Real-time order display and management for kitchen staff.
 - **Manager Web Dashboard:** Comprehensive insights and administrative controls for managers.
- **API Gateway:** Serves as the single entry point for all client requests, handling authentication, routing, and WebSocket connections to various backend services.
- **Core Business Logic Layer (Modular Services):** This layer comprises specialized services that encapsulate business logic, categorized to align with the primary user workflows and pages.
 - **Dashboard Module (Analytics Focus):**
 - **Analytics Service:** Responsible for data aggregation, reporting, and real-time metrics for the Manager Dashboard.
 - **Order Management Module (Core Operations):** This module consolidates services critical for handling orders, tables, billing, and kitchen operations.
 - **Order Service:** Manages order lifecycle, creation, modifications, and status updates.
 - **Menu Service:** Handles menu item management, categories, variants, and pricing rules.
 - **Table Service:** Manages table status, assignments, and reservations for dine-in.
 - **Billing Service:** Calculates bills, applies discounts/taxes, and manages

- payment processing.
- **Kitchen Service:** Manages Kitchen Order Tickets (KOTs), food preparation status, and kitchen display updates.
- **Receipts Module (History & Records):**
 - **Order Service:** Provides historical order data.
 - **Billing Service:** Provides detailed receipt and payment transaction history.
 - **Customer Service:** Manages customer profiles, order history, and loyalty programs.
- **Online Order Module (External Integrations):**
 - **Online Integration Service:** Manages connections and data exchange with third-party online platforms (e.g., Swiggy/Zomato) and the restaurant's own website for online orders.
 - **Menu Service:** Integrates with and updates external platforms' menus and product availability.
- **Supporting Functionalities (Integrated Features/Pop-ups):**
 - **Authentication:** Handled primarily by the API Gateway with an underlying Auth component/service for user and role management, typically presented via login/permission pop-ups.
 - **Printing:** A dedicated Print component/service responsible for generating and routing print jobs (e.g., KOTs, bills), often triggered as a pop-up or background process.
 - **Tax Calculation:** An integrated Tax component/service within the Billing Service that applies complex tax rules, rather than a separate core microservice.

2. Detailed Feature Breakdown (Aligned with Pages)

2.1 Dashboard Page

This page provides an overview of the restaurant's performance and quick actions.

- **Real-time Metrics:**
 - Today's sales, order distribution, live order count, payment breakup.
 - **Technical Implementation:** WebSocket updates, Redis caching, aggregation queries, Chart.js visualization.
- **Quick Actions:**
 - Buttons for new orders, recent orders, pending deliveries.
 - Push notifications for low stock alerts.
 - **Technical Implementation:** Action shortcuts, priority queuing.
- **Analytics Widgets:**

- Sales trends, popular items, peak hours, staff performance.
- **Technical Implementation:** Time-series data (MongoDB aggregation), predictive analytics, export functionality.

2.2 Order Page (Table, Billing, Kitchen Services)

The central hub for taking, managing, and processing orders across different types.

2.2.1 Order Types and Workflows

- **Dine-in:**
 - Table selection, status management, order transfer, split bills.
 - **Technical Requirements:** Real-time table sync, order state machine, WebSocket broadcasting.
- **Takeaway:**
 - Customer management, pickup time slots, queue management.
 - **Technical Requirements:** Customer database, time slot algorithm, SMS gateway integration.
- **Home Delivery:**
 - SMS notifications, address management, delivery zones, driver assignment, route optimization.
 - **Technical Requirements:** Maps integration, geofencing, driver app APIs.

2.2.2 Live KOT Modification System

- **Real-time Editing:**
 - Live quantity updates, item additions/deletions, special instructions.
 - Kitchen notifications.
 - **Technical Implementation:** WebSocket connections, optimistic UI updates, conflict resolution, event sourcing.
- **Version Control:**
 - Change history, audit trail, rollback capability, user attribution.
 - **Technical Implementation:** Event log storage, change tracking.
- **Kitchen Sync:**
 - Instant updates, color-coding changes, alert sounds, acknowledgment system.
 - **Technical Implementation:** Push notifications, queue management, status tracking.

2.2.3 Menu Management

- **Category Management:** Multi-level categories, tax assignment, display ordering, active/inactive status.
 - **Technical Implementation:** Tree structure, tax rule engine, drag-and-drop

ordering.

- **Item Management:** Variants & modifiers, image management, nutritional info, recipe linking.
 - **Technical Implementation:** JSONB attributes, CDN integration, composition pattern.
- **Pricing Engine:** Base pricing, time-based pricing, combo deals, happy hour rules.
 - **Technical Implementation:** Rule engine, temporal tables, price history.
- **Bulk Operations:** Excel import/export, batch updates, category moves, price adjustments.
 - **Technical Implementation:** Excel parsing, background jobs, transaction handling.

2.2.4 Billing and Payment System

- **Bill Generation:** Auto tax calculation, discount application, service charges, rounding rules.
 - **Technical Implementation:** Tax engine, discount rules, calculation precision.
- **Payment Processing:** Multiple payment modes, split payments, partial payments, refund handling.
 - **Technical Implementation:** Payment gateway APIs, transaction logging, reconciliation.
- **UPI Integration:** Dynamic QR codes, payment verification, auto-reconciliation.
 - **Technical Implementation:** UPI gateway APIs, QR generation, webhook handlers.

2.3 Receipt Pages (Order History & Receipts)

This section focuses on retrieving and displaying past orders and their corresponding receipts.

- **Invoice Management:** Custom templates, multi-language support, email/SMS delivery, reprint functionality.
 - **Technical Implementation:** PDF generation, template engine, delivery queue.
- **Order History:** Comprehensive list of all past orders with status, payment details, and items.
 - **Technical Implementation:** Querying Order Service and Billing Service, data presentation.

2.4 Online Order Page

Dedicated interface for managing and fulfilling online orders.

- **Platform Integration (Swiggy/Zomato):** API authentication, order polling, status updates, menu sync.

- **Technical Implementation:** OAuth integration, webhook handling, API rate limiting.
- **Own Website Integration:** Order API, real-time sync, inventory updates.
 - **Technical Implementation:** REST APIs, WebSocket updates.
- **Unified Dashboard:** Aggregated view of all online orders.
- **Order Management:** Auto-acceptance rules, rejection handling, notification system.
 - **Technical Implementation:** Order aggregation, rule engine.